Lec-45

31/08/26 7:12 PM

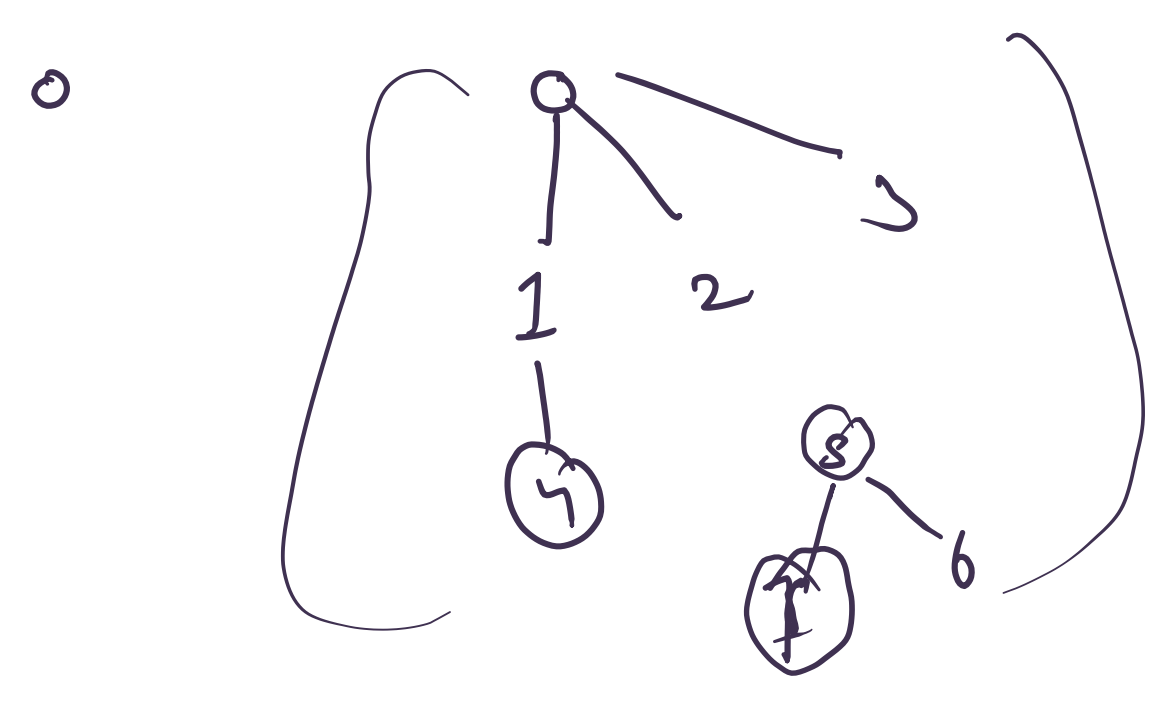
You have a graph of `n` nodes labeled from `0` to `n - 1`. You are given an integer n and a list of `edges` where `edges[i] = [a_i, b_i]` indicates that there is an undirected edge between nodes `a_i` and `b_i` in the graph.

Return `true` *if the edges of the given graph make up a valid tree, and* `false` *otherwise.*

Tree → Graph

Graph → Tree ×

n = 5, edges = {[[0,1],[0,2],[0,3],[1,4]]}



Cycle

Connected 1

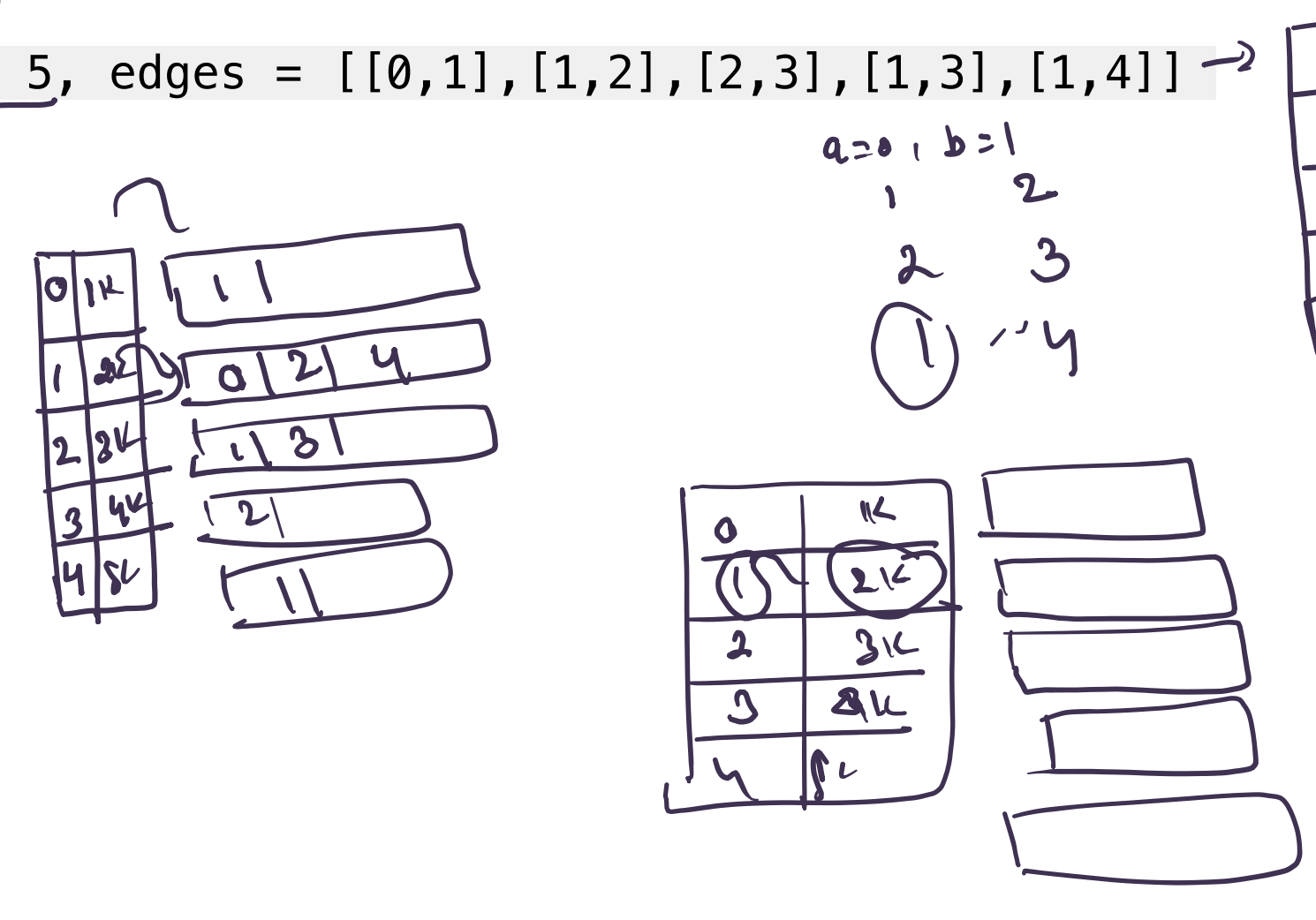
0 1 2 3 4

n = 5, edges = [[0,1],[1,2],[2,3],[1,3],[1,4]]

0	1
0	1
1	2
2	3
1	3
1	4

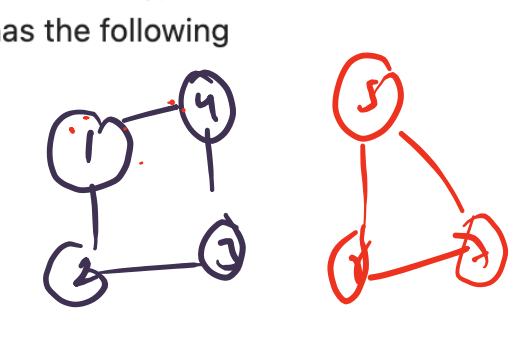
a = edge(i)(0)

b = edge(i)(1)



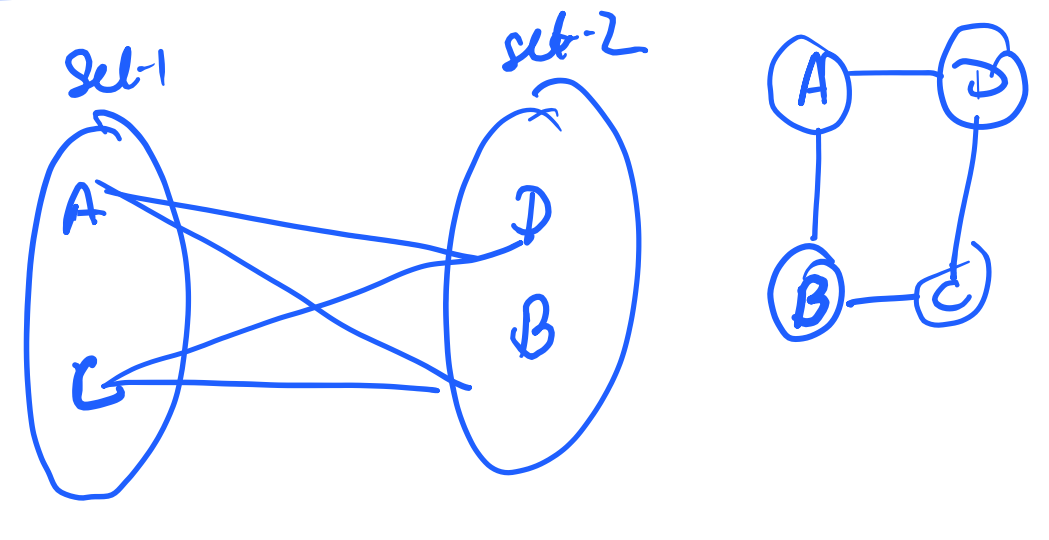
There is an **undirected** graph with `n` nodes, where each node is numbered between `0` and `n - 1`. You are given a 2D array `graph`, where `graph[u]` is an array of nodes that node `u` is adjacent to. More formally, for each `v` in `graph[u]`, there is an undirected edge between node `u` and node `v`. The graph has the following properties:

- There are no self-edges (`graph[u]` does not contain `u`). ✗
- There are no parallel edges (`graph[u]` does not contain duplicate values).
- If `v` is in `graph[u]`, then `u` is in `graph[v]` (the graph is undirected).
- The graph may not be connected, meaning there may be two nodes `u` and `v` such that there is no path between them.

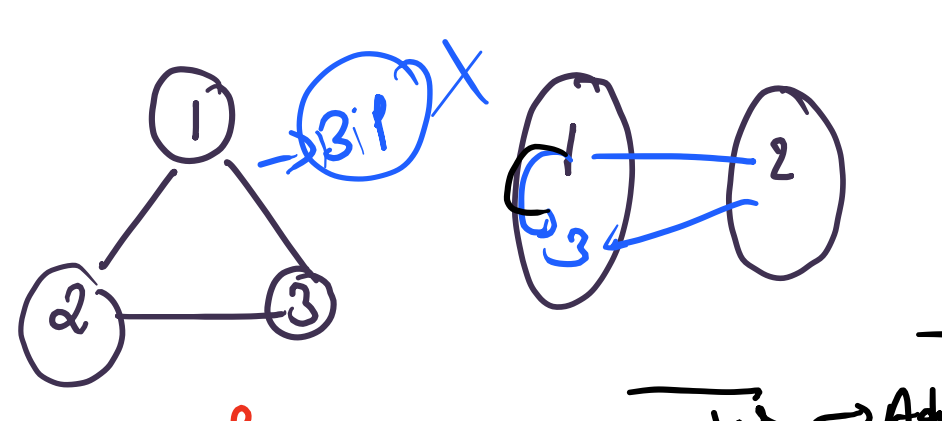


A graph is **bipartite** if the nodes can be partitioned into two independent sets `A` and `B` such that **every** edge in the graph connects a node in set `A` and a node in set `B`.

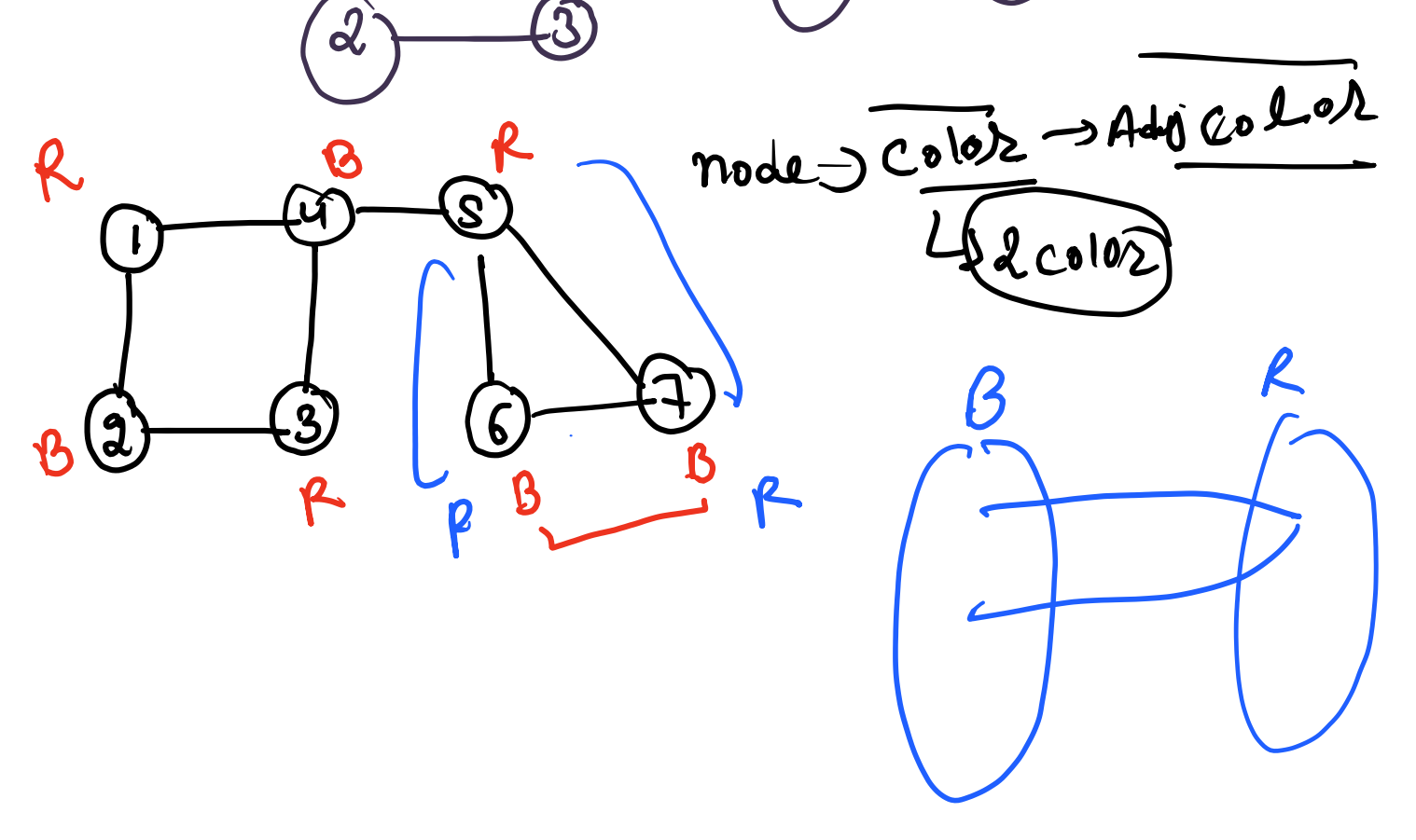
Return `true` *if and only if it is* ***bipartite***.



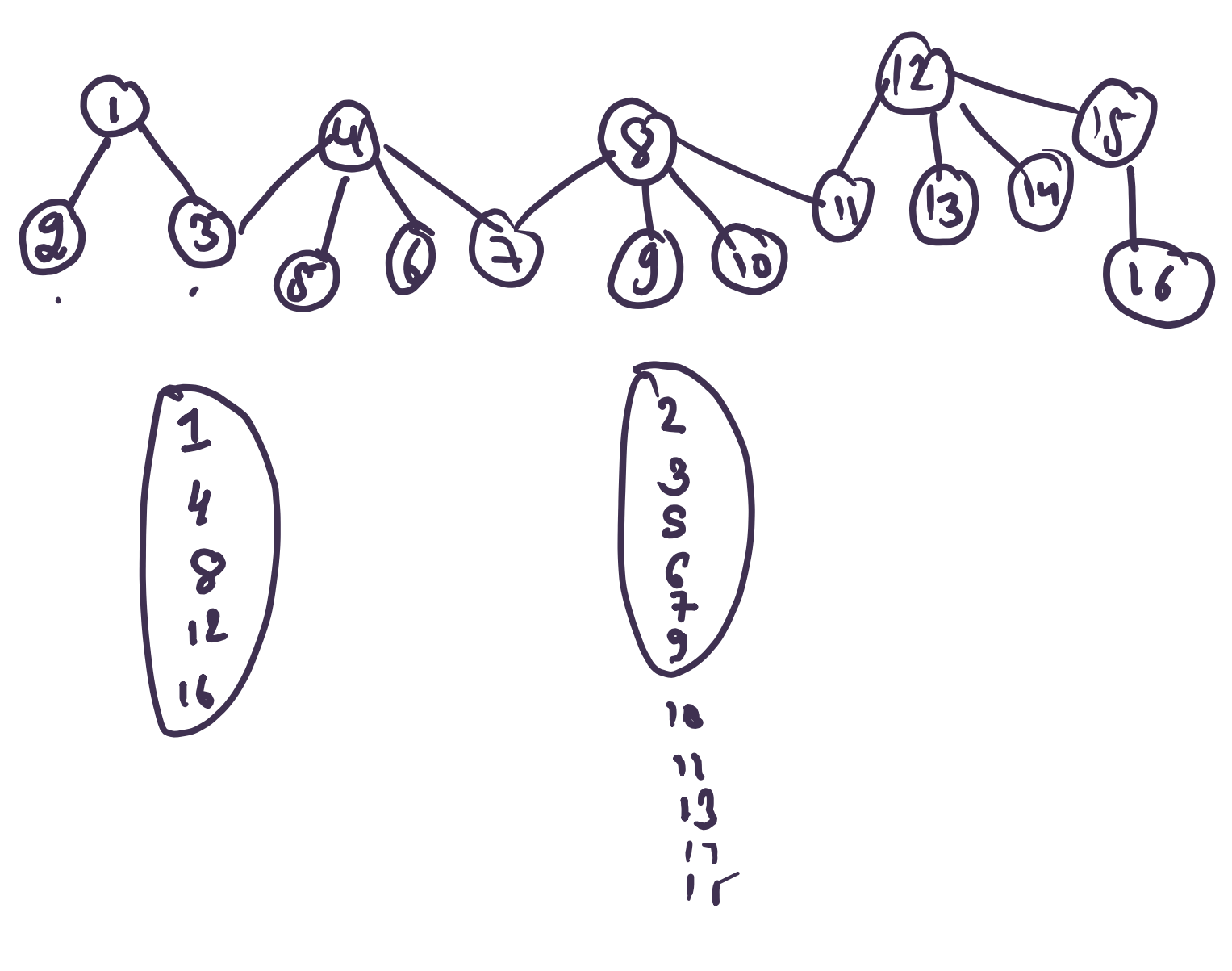
① Bip



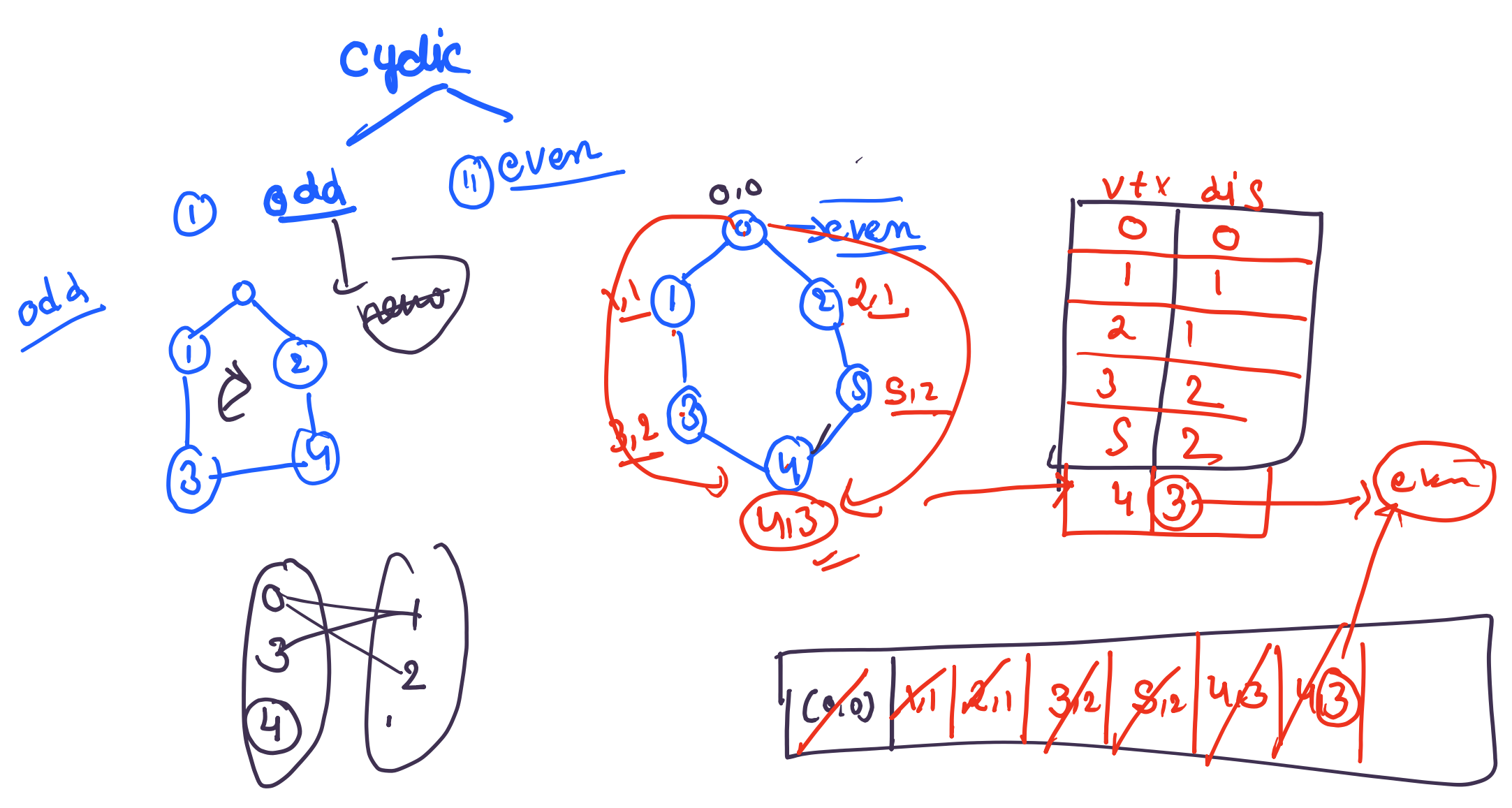
(ii)



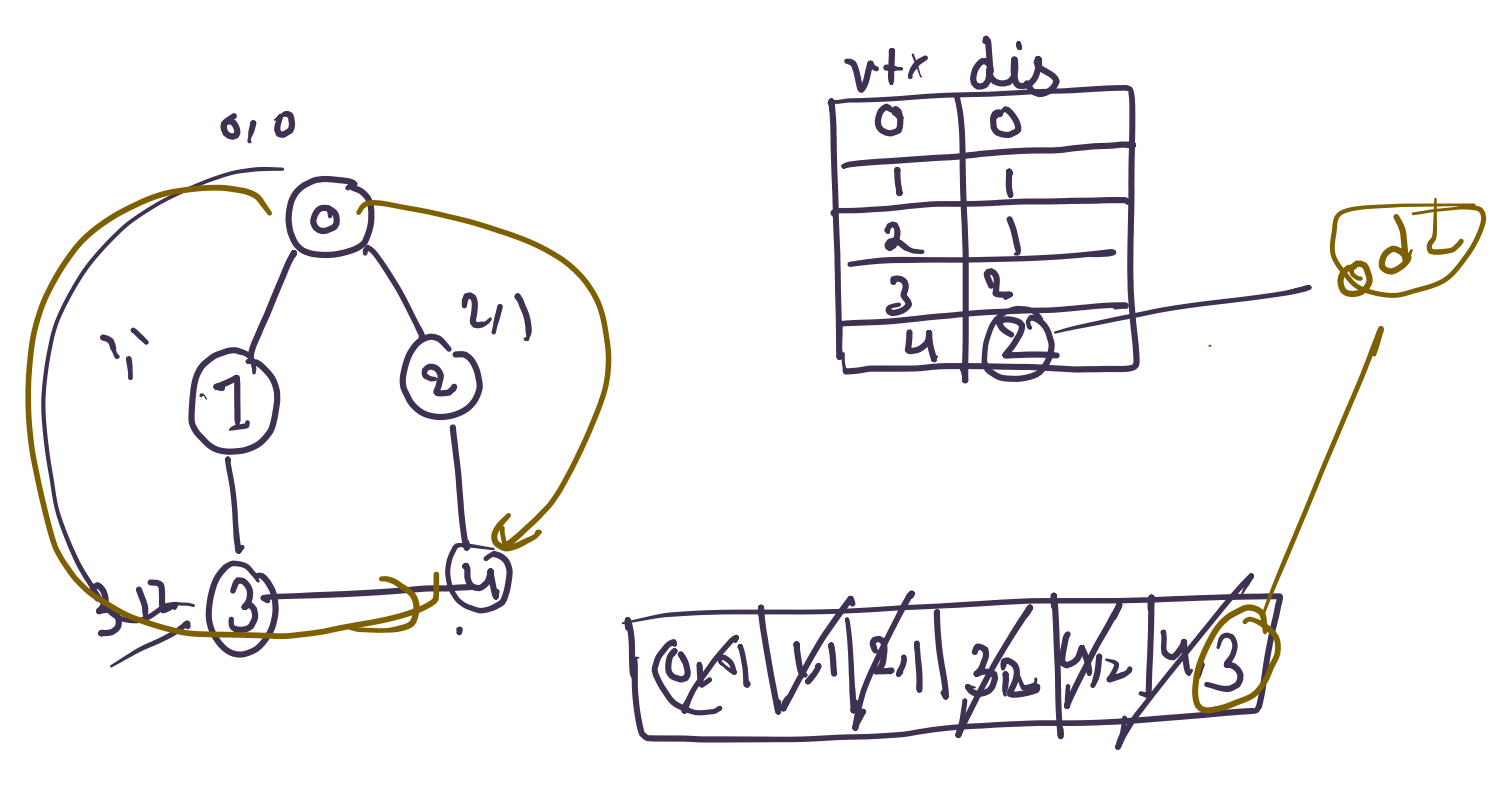
① Cyclic ⓘⓘ acyclic → Bi Partite



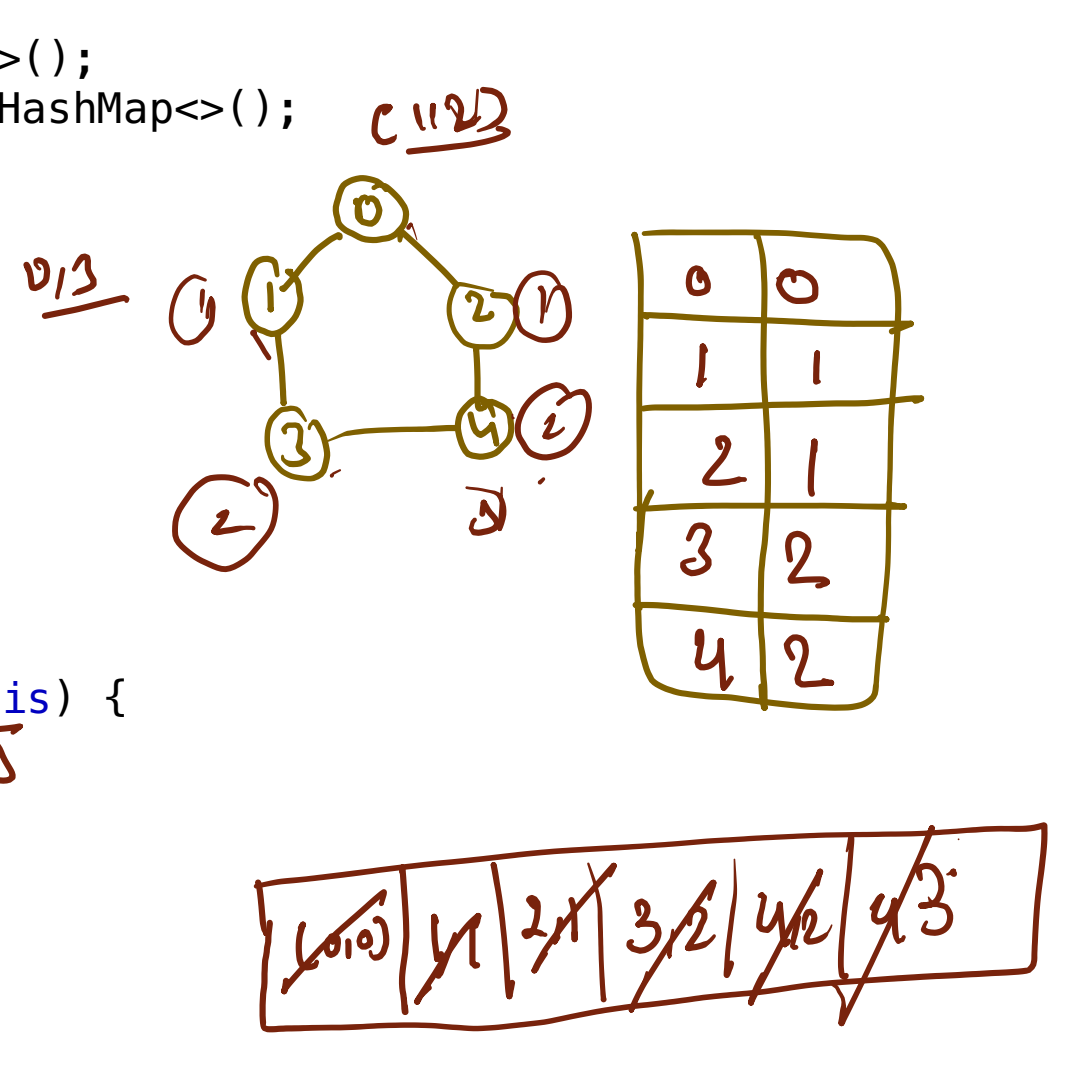
Cyclic



1. remove
2. Ignore
3. visited mark
4. Self work
5. Add nbrs

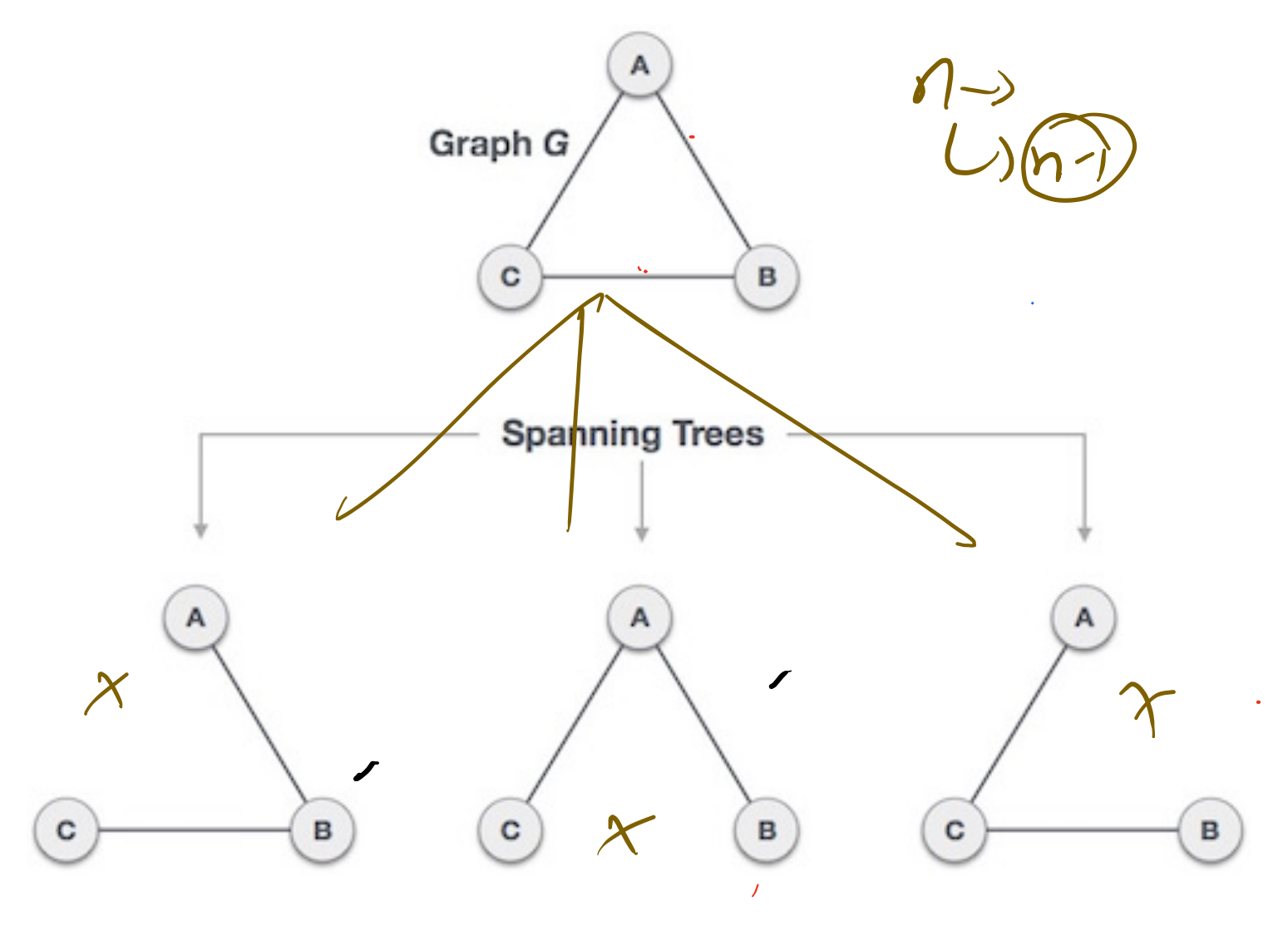


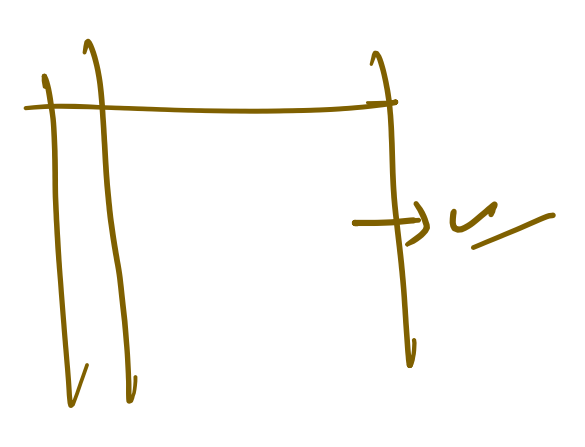
```
public boolean isBipartite(int[][] graph) {
        Queue<BipartitePair> q = new LinkedList<>();
        HashMap<Integer, Integer> visited = new HashMap<>();
        for (int i = 0; i < graph.length; i++) {
            if (visited.containsKey(i)) {
                continue;
            }
            // BFS;
            q.add(new BipartitePair(i, 0));
            while (!q.isEmpty()) {
                // 1. remove
                BipartitePair rp = q.poll();
                // 2. Ignore if Already visited
                if (visited.containsKey(rp.vtx)) {
                    if (visited.get(rp.vtx) != rp.dis) {
                        return false;
                    }
                    continue;
                }
                // 3. Marked visited
                visited.put(rp.vtx, rp.dis);
                // 4. self work
                // 5. Add unvisited nbrs
                for (int nbrs : graph[rp.vtx]) {
                    if (!visited.containsKey(nbrs)) {
                        q.add(new BipartitePair(nbrs, rp.dis + 1));
                    }
                }
            }
        }
        return true;
    }
```



Spanning Tree Subset

A spanning tree is a subset of Graph G, which has all the vertices covered with minimum possible number of edges. Hence, a spanning tree does not have cycles and it cannot be disconnected..





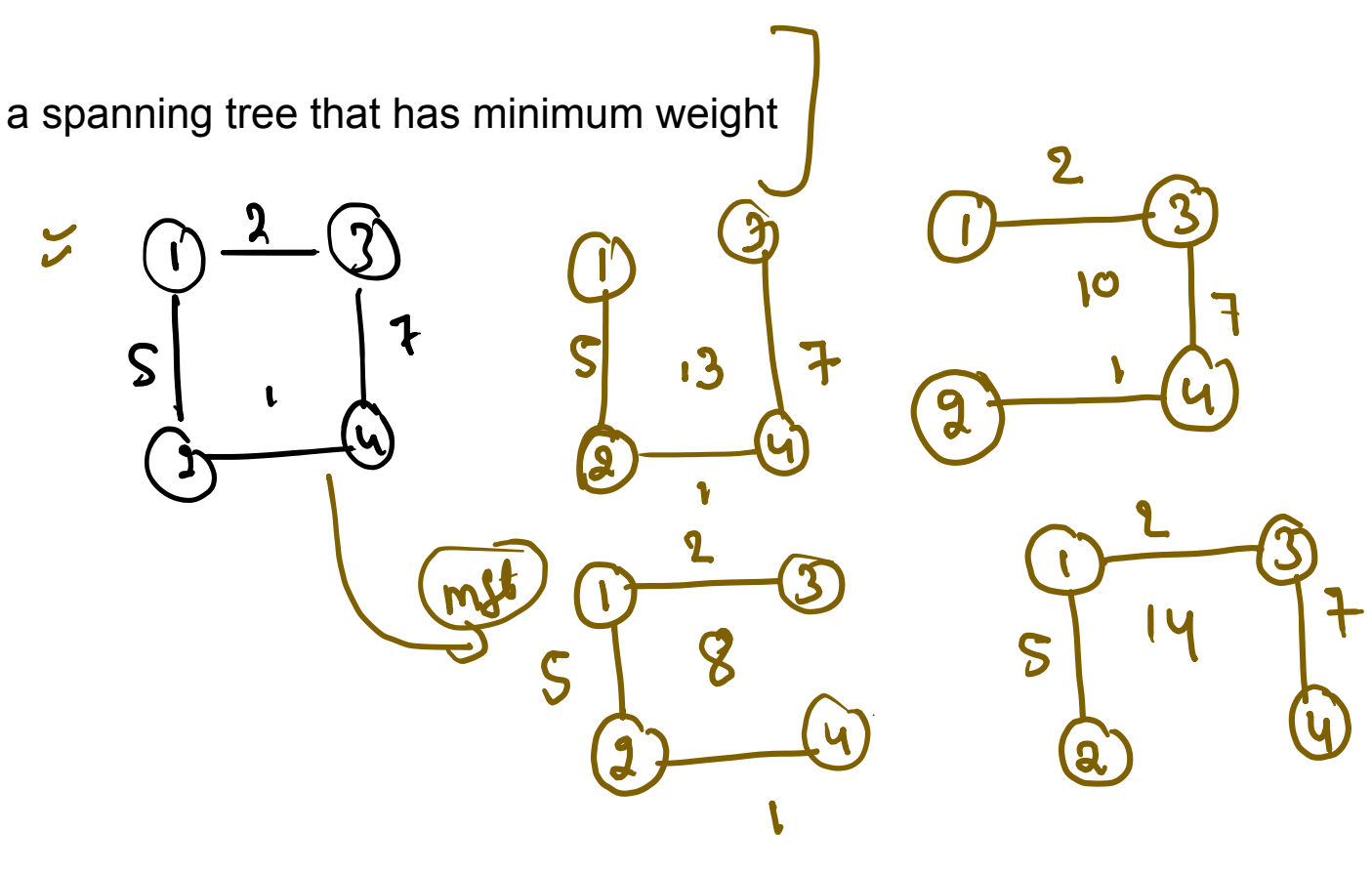
Properties of Spanning Tree

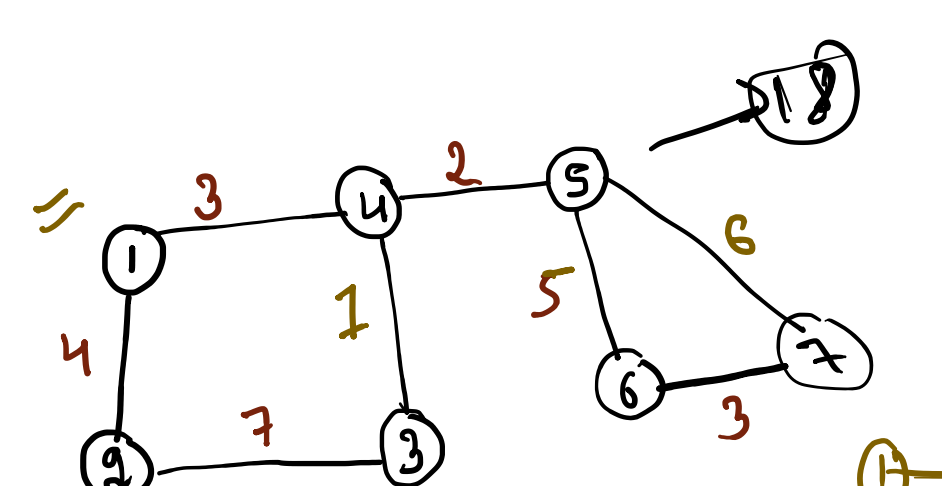
- A connected graph G can have more than one spanning tree.
- Spanning Tree has n-1 edges, where n is the number of Node (Vertices)
- All possible spanning trees of graph G, have the same number of edges and vertices.
- The spanning tree does not have any cycle (loops).
- Removing one edge from the spanning tree will make the graph disconnected, i.e. the spanning tree is minimally connected.
- Adding one edge to the spanning tree will create a circuit or loop, i.e. the spanning tree is maximally acyclic.

Minimum Spanning Tree (MST)
In a weighted graph, a minimum spanning tree is a spanning tree that has minimum weight than all other spanning trees of the same graph.

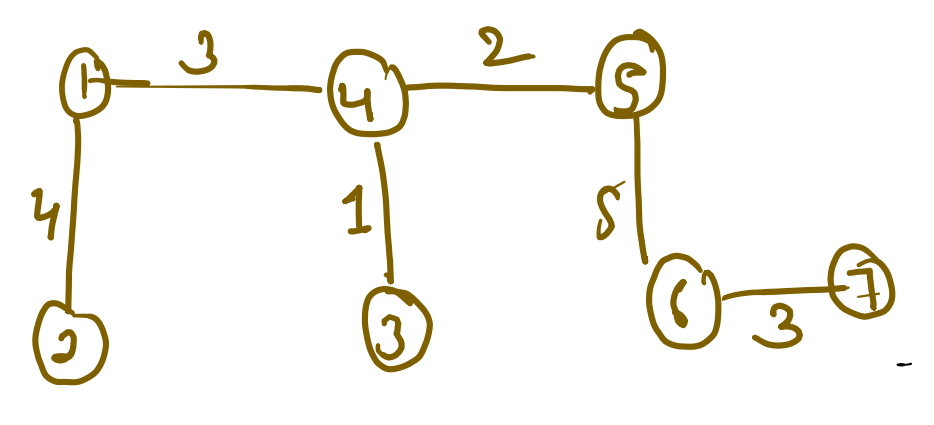
- Kruskal's Algorithm

Prim's Algorithm
Both are greedy algorithms.

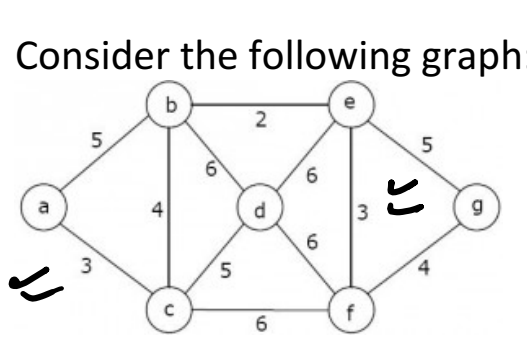




1+2+3+3+4+5



Consider the following graph:



Which one of the following is NOT the sequence of edges added to the minimum spanning tree using Kruskal's algorithm?

(A) (b,e)(e,f)(a,c)(b,c)(f,g)(c,d)
(B) (b,e)(e,f)(a,c)(f,g)(b,c)(c,d)
(C) (b,e)(a,c)(e,f)(b,c)(f,g)(c,d)
(D) (b,e)(e,f)(b,c)(a,c)(f,g)(c,d) ✗

3 4 3

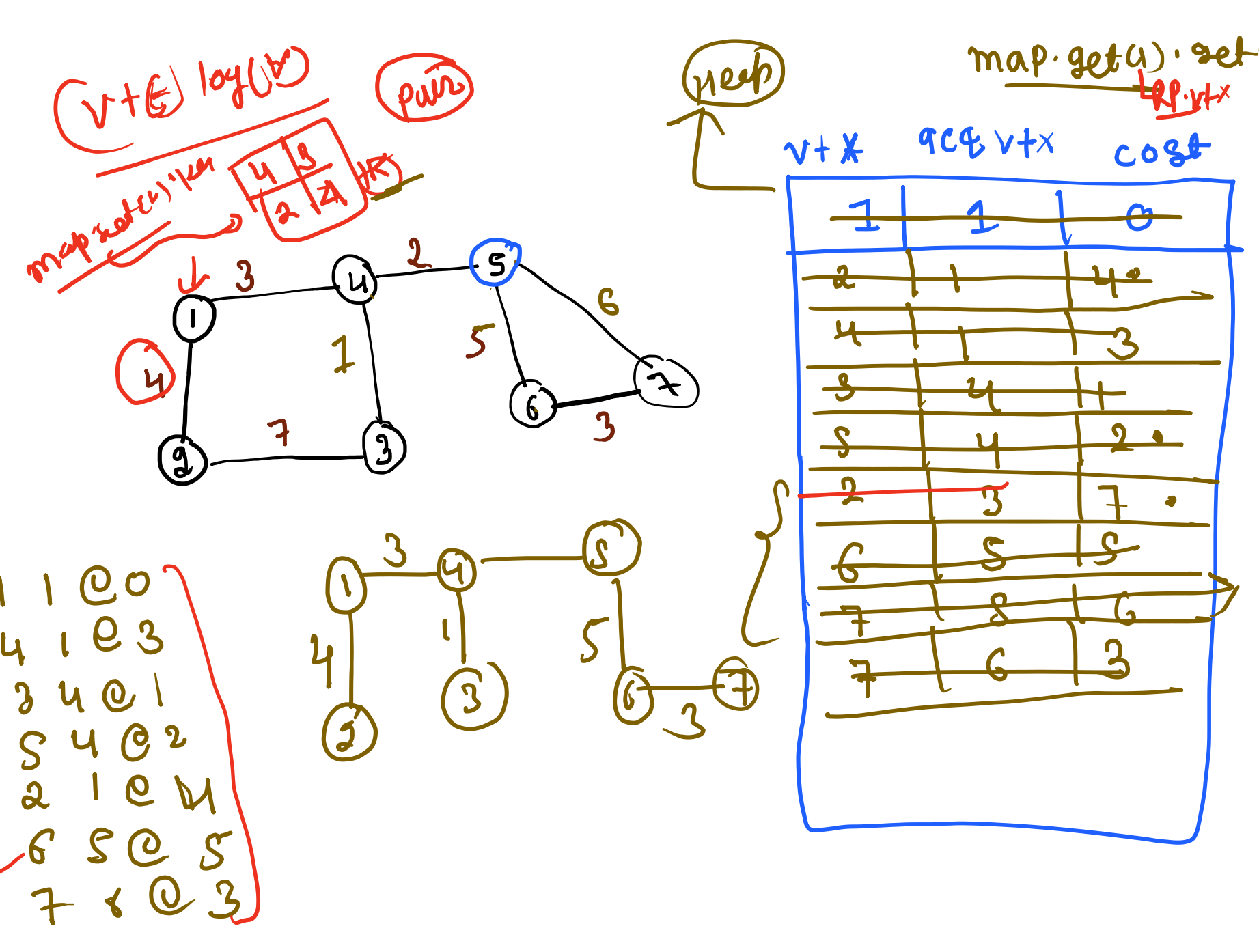
Complexity

Kruskal → C.P → D.S.U

Prims

Big-calc

1. Remove
2. Ignore
3. visited
4. self work
5. Add nbrs

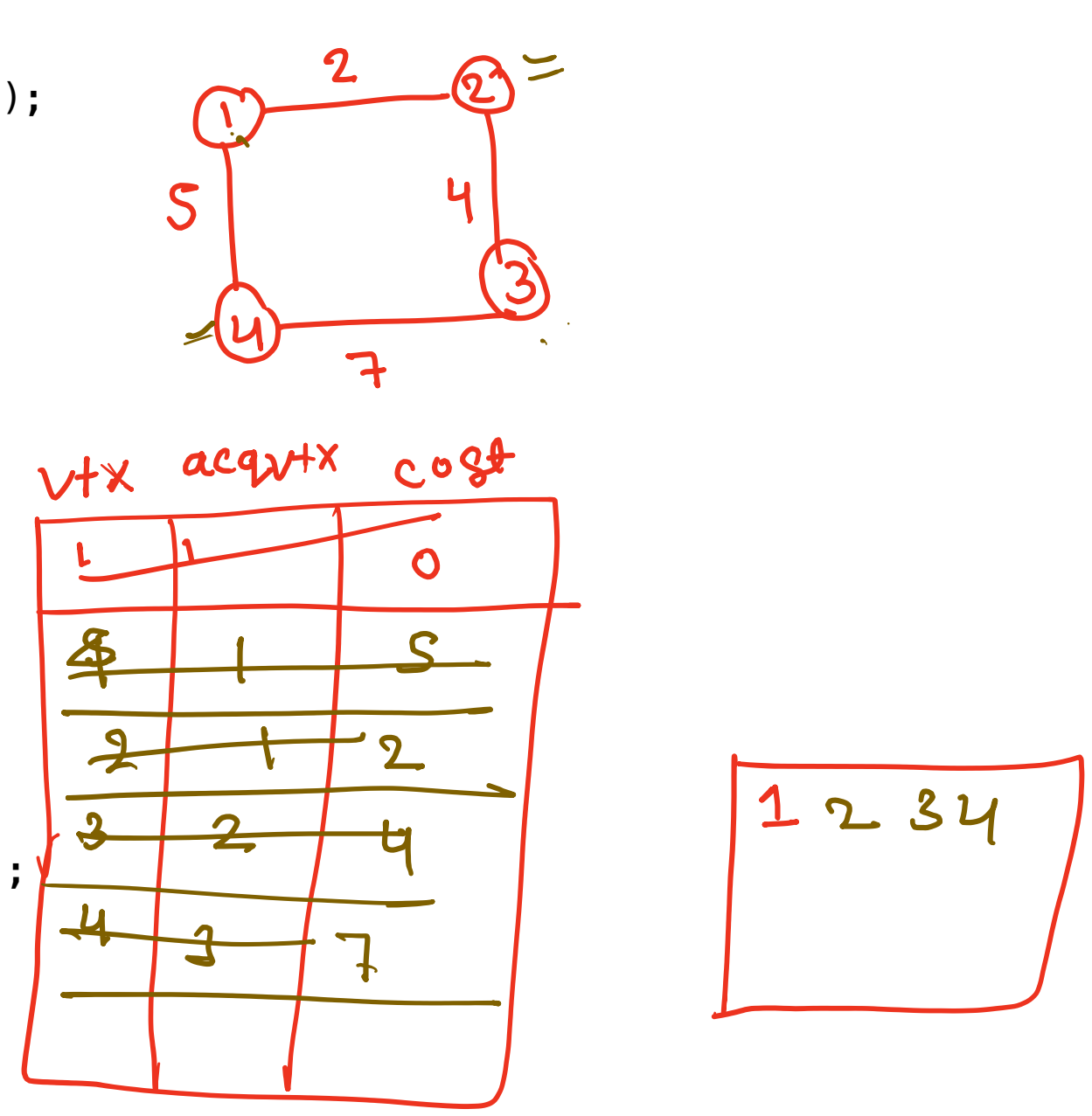


1 4 3 5
2 6 7

1. Remove
2. Ignore
3. visited
4. self work
5. Add nbrs

2, 1, 4

```
public void PrimsALgo() {
        PriorityQueue<PrimsPair> pq = new PriorityQueue<>();
        HashSet<Integer> visited = new HashSet<>();
        pq.add(new PrimsPair(1, 1, 0));
        while (!pq.isEmpty()) {
            // 1. remove
            PrimsPair rp = pq.poll();
            // 2. Ignore if Already visited
            if (visited.contains(rp.vtx)) {
                continue;
            }
            // 3. Marked visited
            visited.add(rp.vtx);
            // 4. Self Work
            System.out.println(rp);
            // 5. Add unvisited nbrs
            for (int nbrs : map.get(rp.vtx).keySet()) {
                if (!visited.contains(nbrs)) {
                    int cost = map.get(rp.vtx).get(nbrs);
                    pq.add(new PrimsPair(nbrs, rp.vtx, cost));
                }
            }
        }
    }
```

1 1 @ 0
2 1 @ 2
3 2 @ 4
4 1 @ 5